SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

I [illegible] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 6marks	Q2 6marks	Q3 6marks	Q4 6marks	Q5 6marks
Problem Identification	20%	1	6	1	1	1
Debugging	25%	1	1	1	1	1
Optimization	20%	2	2	1	2	2
Analysis	20%	1	1	1	1	1
Code Quality	15%	0	1	6	0	0

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Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

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Problem Identification	20%				
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Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A: Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:
IP Address: 192.168.10.65
Subnet Mask: 255.255.255.192 (/26)
Default Gateway: 192.168.10.1
However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task
Debug the network configuration by:
- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.

- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:
Device A: 2001:db8::ff00:42:8329/64
Device B: 2001:db8:0:0:1::1/64
Although both devices are connected to the same network, communication fails.

Task
Debug the IPv6 configuration by:
- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task
Analyze and optimize the subnet allocation by:
- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:
Device A: 2001:db8::ff00:42:8329/64
Device B: 2001:db8:0:0:1::1/64
Although both devices are connected to the same network, communication fails.

Task
Debug the IPv6 configuration by:
- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.
Destination Network: 192.168.1.0/24, Next Hop: 10.0.0.2

Destination Network: 192.168.2.0/24, Next Hop: 10.0.0.3
Default Route, Next Hop: 10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Task
Debug the routing configuration by:
- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-	Code is mostly clear	Code lacks structure and	Poorly written code

		structured code with proper comments and documentation	with minor documentation gaps	proper documentation	with no documentation

Answers to Assessment

1. Debugging IPv4 Workstation Configuration

Given:
- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1
- Intended subnet: 192.168.10.0/26

Step 1: Identify the subnet

A /26 subnet has a block size of: 256 - 192 = 64
The subnet ranges are 0–63, 64–127, 128–191, and 192–255.
Therefore, IP address 192.168.10.65 belongs to subnet 192.168.10.64/26, not 192.168.10.0/26.

Step 2: Determine the subnet details

Parameter	Correct Value
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
First Valid Host	192.168.10.65
Last Valid Host	192.168.10.126
Total IP Addresses	64
Usable Host Addresses	62

Step 3: Verify the IP address and gateway

- IP address: 192.168.10.65 is valid in the 192.168.10.64/26 subnet.
- Gateway: 192.168.10.1 belongs to 192.168.10.0/26, so it is outside the workstation's subnet.

Step 4: Identify the configuration error

The workstation's IP address and default gateway belong to different subnets. The workstation is also configured outside the intended 192.168.10.0/26 network.

Step 5: Correct the configuration

To use the intended subnet, configure:

Setting	Correct Configuration
IP Address	192.168.10.2
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.1
Network Address	192.168.10.0
Broadcast Address	192.168.10.63

The corrected subnet supports: 2^(32-26) - 2 = 62 usable hosts.
Conclusion: Assign an IP address in the 192.168.10.0/26 range and use gateway 192.168.10.1. The gateway must also be configured and reachable on that subnet.

2. Debugging IPv6 Address Configuration

Given:
- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Step 1: Expand both IPv6 addresses

Device A — full notation:
2001:0db8:0000:0000:0000:ff00:0042:8329/64
Device B — full notation:
2001:0db8:0000:0000:0001:0000:0000:0001/64

Step 2: Determine network prefixes

A /64 prefix means the first four 16-bit blocks form the network prefix.

Parameter	Device A	Device B
IPv6 Address	2001:db8::ff00:42:8329	2001:db8:0:0:1::1
Prefix Length	/64	/64
Network Prefix	2001:db8:0:0::/64	2001:db8:0:0:1::/64

Step 3: Check whether both devices belong to the same subnet

- Device A belongs to 2001:db8:0:0::/64.
- Device B belongs to 2001:db8:0:0:1::/64.

Result: They belong to different /64 subnets. Their first four hexadecimal blocks are not identical.

Step 4: Identify the error and correct it

The addresses use different network prefixes. If both devices are intended to communicate directly on the same /64 LAN, assign Device B an address in Device A's prefix.
For example:

Setting	Corrected Configuration
Device A	2001:db8::ff00:42:8329/64
Device B	2001:db8::1/64
Shared Network Prefix	2001:db8:0:0::/64

Step 5: Calculate host address capacity

IPv6 contains 128 bits. With a /64 prefix:
128 - 64 = 64 host bits
2^64 = 18,446,744,073,709,551,616
Conclusion: Both devices should use addresses within the same /64 prefix for direct communication on the same LAN. The subnet has 2^64 possible interface identifier values.

3. Optimizing IPv4 Subnet Allocation Using VLSM

Given network: 192.168.1.0/24
The network is divided using /26, /27, and /28 subnet masks.

Step 1: Calculate subnet sizes

Prefix	Subnet Mask	Total IPs	Usable Hosts
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14

Step 2: Example of a valid subnet allocation

Since the question does not specify the exact subnet addresses or department requirements, the following is one possible allocation:

Subnet	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
/26	192.168.1.0	192.168.1.1–192.168.1.62	192.168.1.63	62
/27	192.168.1.64	192.168.1.65–192.168.1.94	192.168.1.95	30
/28	192.168.1.96	192.168.1.97–192.168.1.110	192.168.1.111	14

These allocations do not overlap.

Step 3: Calculate unused addresses

The original /24 network contains 256 IP addresses.
Allocated addresses: 64 + 32 + 16 = 112
Remaining addresses: 256 - 112 = 144
The unused range is 192.168.1.112–192.168.1.255, containing 144 addresses.

Step 4: Identify inefficiencies and errors

- A department with 20 devices may use a /26, wasting capacity because a /27 provides 30 usable hosts.
- A department with 10 devices can use a /28, providing 14 usable hosts.
- Overlapping subnets or incorrect network boundaries can cause communication failures.
- A /28 cannot support 20 devices because it provides only 14 usable host addresses.

Unused capacity cannot be determined precisely for each department without the actual number of devices and current subnet assignments.

Step 5: Recommend VLSM optimization

Variable Length Subnet Masking (VLSM) allocates different subnet sizes based on each department's requirements.

Required Hosts	Recommended Prefix	Usable Hosts
Up to 14	/28	14
Up to 30	/27	30
Up to 62	/26	62
Up to 126	/25	126

Conclusion: Allocate the largest required subnet first, followed by smaller subnets. This reduces address wastage and prevents overlap. The exact optimized design depends on the real host requirements of each department.

4. Debugging IPv6 Address Configuration (Repeated Question)

This question is identical to Question 2.

Step 1: Full IPv6 notation

- Device A: 2001:0db8:0000:0000:0000:ff00:0042:8329/64
- Device B: 2001:0db8:0000:0000:0001:0000:0000:0001/64

Step 2: Compare network prefixes

Device	Network Prefix
Device A	2001:db8:0:0::/64
Device B	2001:db8:0:0:1::/64

Both devices belong to different /64 subnets.

Step 3: Correct configuration

Use the same prefix for both devices:
- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8::1/64

Shared prefix: 2001:db8:0:0::/64

Step 4: Host address capacity

2^(128-64) = 2^64

Total possible interface identifier values: 18,446,744,073,709,551,616.

Conclusion: Both devices must use addresses from the same /64 subnet for direct communication on the same LAN.

5. Debugging Router Routing Table

Given routing table:

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP: 192.168.2.100

Step 1: Identify the matching route

The destination IP 192.168.2.100 belongs to network 192.168.2.0/24.

Therefore, the matching routing entry is: 192.168.2.0/24 → 10.0.0.3

Step 2: Determine the next-hop router

The router selects next hop 10.0.0.3.
This route is preferred over the default route because /24 is a more specific match than /0. This is known as longest-prefix matching.

Step 3: Verify the destination subnet

For a /24 network:
- Network address: 192.168.2.0
- Valid host range: 192.168.2.1–192.168.2.254
- Broadcast address: 192.168.2.255

The destination 192.168.2.100 is a valid host address in this subnet.

Step 4: Identify possible routing errors

If packets do not reach the destination, check:
1. Whether next hop 10.0.0.3 is reachable.
2. Whether the next-hop router has a valid route to 192.168.2.0/24.
3. Whether the destination host is active and configured correctly.
4. Whether firewalls or access control lists block traffic.
5. Whether the return path is configured correctly.

The given table alone does not prove which of these errors is present.

Step 5: Optimize the routing table

Ensure that the specific route is configured correctly:

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
0.0.0.0/0	10.0.0.1

The table already contains a specific route and a default route. No change is necessary unless troubleshooting confirms an incorrect or unreachable next hop.

Step 6: Identify the default route

If no more specific route matches the destination, the router uses:
0.0.0.0/0 → 10.0.0.1
This route is used only if it is installed and available.
Conclusion: The destination 192.168.2.100 matches 192.168.2.0/24, so the router forwards packets to next hop 10.0.0.3. If delivery fails, verify next-hop reachability, downstream routing, and the return path.